EEE3308C: Electronic Circuits 1

Lab 4: MOSFET Amplifiers – Pre-Lab

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Objectives

- Understand how MOSFETs work and their properties.
- Introduction to the design and operation of a common-source (CS) amplifier.

Equipment

- 2 each: 2N7000 NMOS FET
- Multiple capacitors
- Various resistors, ranging from 100 to 10k ohms
- Breadboard and various wires
- Digilent Analog Discovery (DAD) Board

Introduction

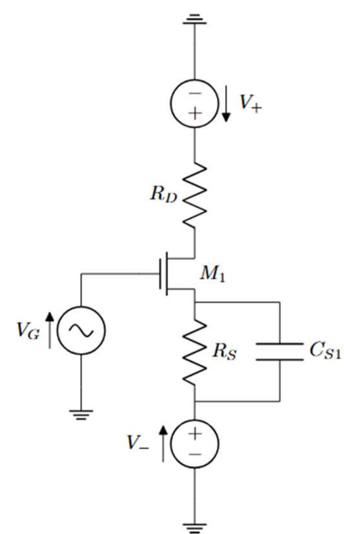


Figure 1 – MOSFET amplifier

- **Drain Resistance**
  In this lab we will manipulate the drain resistor (R_D=220 Ω) in circuits like the one above to perform certain functions. For this lab, this includes measuring drain current (I_D) and serving as the load resistor in a common-source amplifier.
- **Testing/Tools**
  We will test the circuits two ways:
  - With the waveform generator as a DC voltage source
  - With a sine wave superimposed on a DC bias voltage.

  We will use the scope in two ways as well:
  - As a DC voltmeter
  - In XY mode to plot V_{out} vs. V_{in}

- **Structure**
  We will first characterize the current-voltage behavior of your MOSFET and then choose a good operating point for the circuit as an amplifier. Then we will characterize the circuit as an amplifier. Finally, we will add a circuit using a negative power supply to set a more stable bias current and characterize the improved circuit. All the circuits will be built using variations on same setup in figure 1.

- **Circuit Construction Tips/Notes:**
  - The MOSFET you will be using is a **2N70000**
  - Be sure you are correctly placing the MOSFET in the breadboard as many people flip-flop the drain and source terminals when building circuits.
  - Use WaveGen 1 for V_G and set the V_+ supply to +5 V.
  - Connect Channel 1 of the scope to measure V_G as shown.
  - Connect Channel 2 of the scope as shown to measure V_D.
  - Set the scope to automatic repeated triggering mode so you can see both traces even with zero signal.
  - Use the **Basic** rather than the **Simple** interface
  - Set WaveGen 1 into DC mode, initially with 0 VDC output.

*Note that $V_D = V_+ - I_D \cdot R_D$. With $V_G = 0$, you should see 0 V across R_D because the MOSFET requires V_{GS} > the threshold voltage V_T for current to flow. So, you should see $V_D = V_+ = 5.0$ V.

Pre-Lab

Part 1: MOSFET Characterization/Modeling (30 Points)

I. Build the following circuit in Figure 2 on your breadboard using a 2N7000 Transistor and insert a picture of this. **(5 Points)**

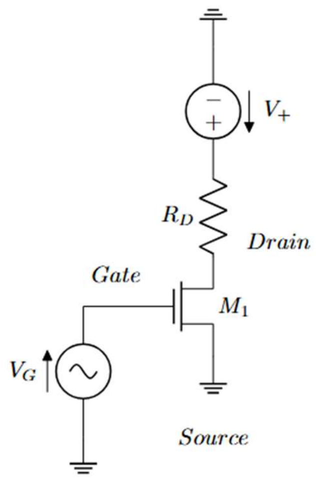


Figure 2 – Simplified MOSFET Test Circuit

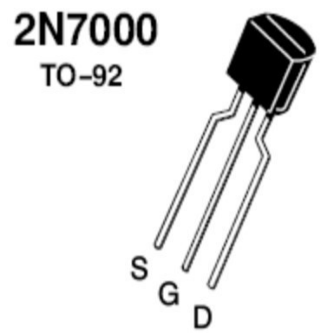
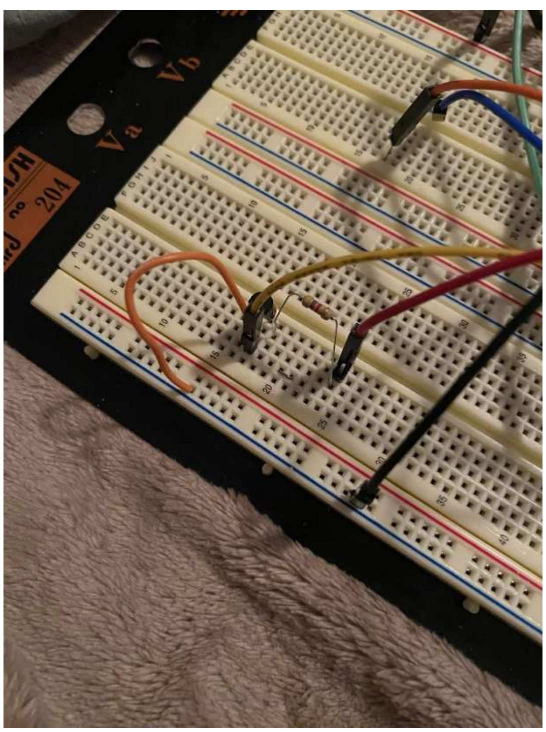


Figure 3 – 2N7000 Pinout



II. Use the slider on the WaveGen to sweep V_G gradually from 0V to 4V. As V_G increases, eventually V_D will begin to drop, indicating some current flow. Estimate the threshold voltage V_T as the value of V_G that just causes V_D to begin to drop. $V_+ = 5$ V. **(2 Points)**

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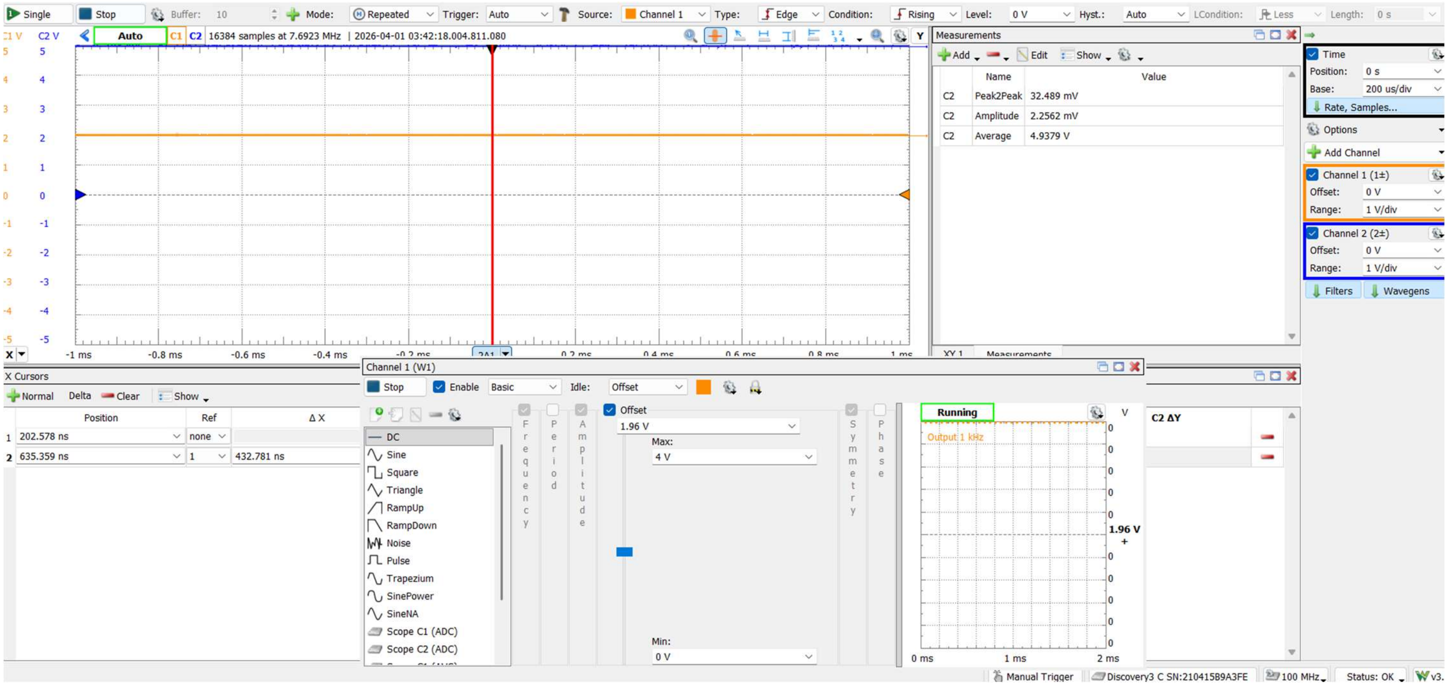


V_T = ~ 1.95V

III. Increase V_G above V_T until V_D drops to about 3.5V.

a. Record the values of V_G and V_D and calculate the corresponding value of $I_D = (5-V_D)/R_D$.

2.26V for VG and 3.4862V for VD → ID = (5-3.4862)/1k = 1.5138mA

b. Determine whether the FET is in the active (i.e. saturation) region or linear. *Hint: $V_{DS} > V_{GS} - V_T$ is in the active region; $V_{DS} < V_{GS} - V_T$ is in the linear region. **(5 Points)**

VGS = 2.26, VDS = 3.4862, VT = 1.95

VDS > VGS – VT → Active

IV. Repeat the above step and measurements, calculations and comparisons for: $V_D \cong 3.0V$, $V_D \cong 2.5V$, $V_D \cong 2.0V$, $V_D \cong 1.5V$, $V_D \cong 1V$, and $V_D \cong 0.5V$. You may use table 1 below for recording the following calculations. **(5 Points)**

V_G (V)	V_D (V)	I_D (A)	$V_D > V_G - V_T$?

2.295	3	2mA	Yes
2.325	2.5	2.5mA	Yes
2.35	2	3mA	Yes
2.375	1.5	3.5mA	Yes
2.398	1	4mA	Yes
2.422	.5	4.5mA	Yes/No (close)

Table 1 - MOSFET

V. What V_G value does V_D go below $V_{Dsat} = V_G - V_T$ (putting the FET into the triode/linear region)? **(3 Points)**

2.42V

VI. Using your estimated value of V_T and one of your data points for which $V_D > V_G - V_T$, estimate the transconductance parameter k using the formula:

$$I_D = {}^{k}/_{2} * (V_{GS} - V_T)^2$$

*This formula is valid throughout the active region for the FET ($V_D > V_G - V_T$). **(2 Points)**

0.003 = k/2 (0.1849)

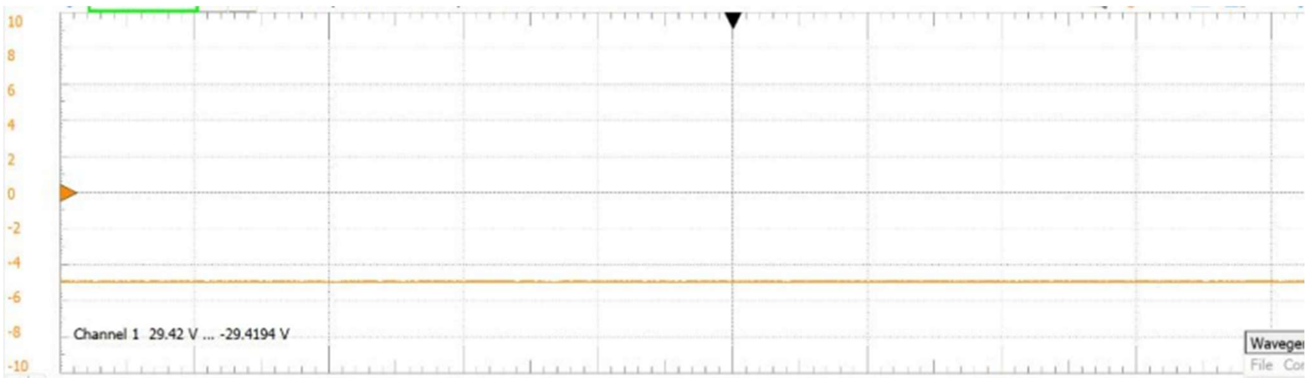
K = 0.003/0.09245 = 0.0324 A/V^2

VII. Plug in a different one of your active-region V_G data points into the square law equation using your estimated value of *k* and V_T (from II. and VI.) and calculate the new I_D. Compare the measured value of I_D to the calculated value. Don't be surprised if the results don't match. This square-law model can be inaccurate at times. **(3 Points)**

$0.478^2 = 0.228$

ID = k2(Vov)^2 = 0.0162 x 0.228 = 3.7mA

VIII. Now we are going to find the same parameters for your PMOS BS250P transistors. Be sure to switch the orientation of the MOSFET and apply the proper voltage (negative voltage) across the gate. Find the following parameters: V_T and k. You do not need to repeat the same steps as above to find the parameters but include screenshots/data to prove your findings. **(5 Points)**



Drop began around -2.45V, so V_T = -2.45V

ID = 0.006 = k/2(VGS-VT)^2 = k/2(-2.05)^2 => k = 0.00286 A/V^2

Part 2: MOSFET Characterization/Modeling - Voltage Transfer Curve (20 Points)

The following graph in figure 4 is a Voltage Transfer Curve (VTC). You will obtain and analyze a VTC for your FET by conducting the following steps. Use the same circuit you initially had in Part 1.

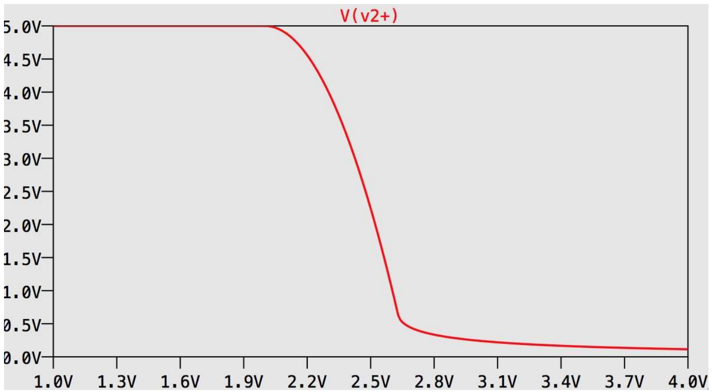
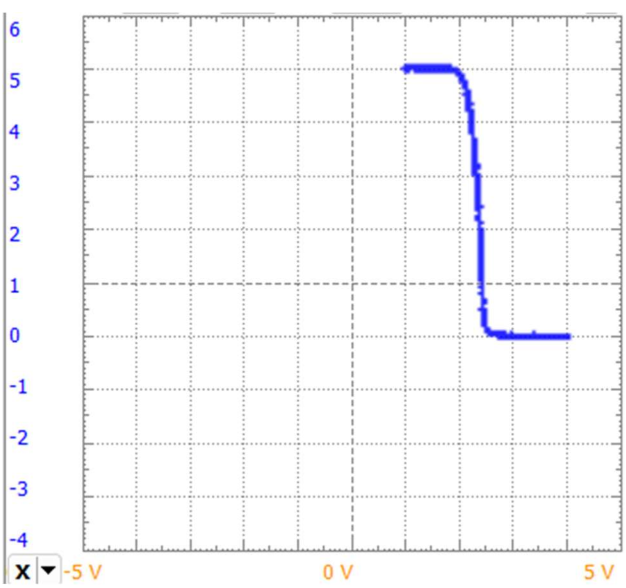


Figure 4 – Typical Voltage Transfer Curve

I. Set up WaveGen1 to supply a sinusoidal function of the following parameters: 2.5V offset, 1.5 V amplitude, 10 kHz frequency. Set up the scope in XY mode with Channel 1 (V_G) on the x-axis and Channel 2 (V_D) on the y-axis. Play with the input offset and amplitude and the scope sensitivities to get a display similar to figure 4. This is the voltage transfer curve (VTC), take a screenshot and insert it. **(5 Points)**



II. Identify the three main regions for the VTC by annotating the figure:

1. For $V_G < V_T$: The FET is in cutoff, with $I_D = 0$ and $V_D = V+$
2. For $V_G > V_D - V_T$, The FET is in the triode region.
3. In between: The FET is in the active region or saturation, where the maximum gain (slope) occurs.

(5 Points)

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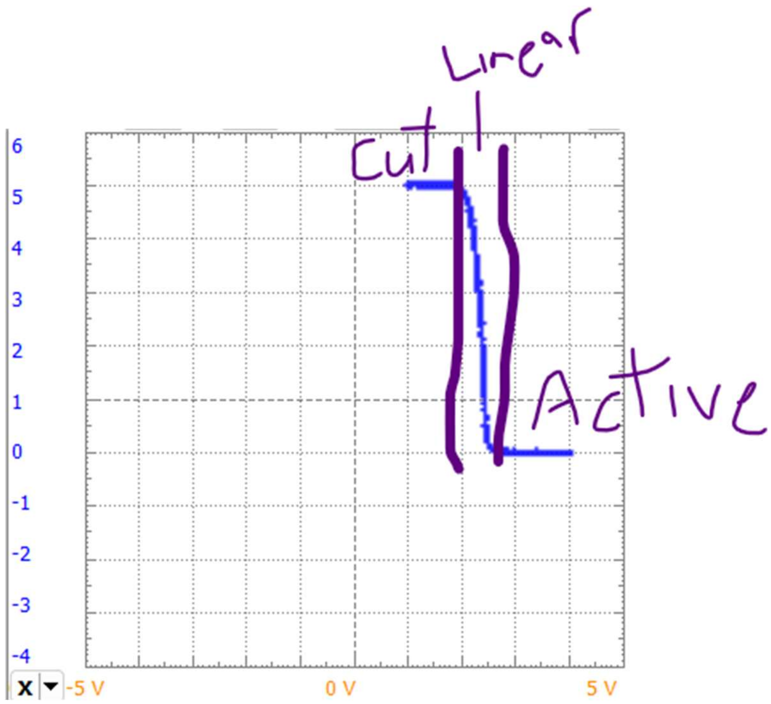

III. Is the VTC a straight line in the active region? **(5 Points)**

It is quite straight once saturated

IV. Repeat the process with your PMOS, be sure to adjust the circuit/MOSFET to obtain a proper VTC. * Note: your will also want to adjust the WaveGen1 input to be in a negative range, choose a range that provides optimal coverage of the graph. Include a screenshot and label the different regions again. **(5 Points)**

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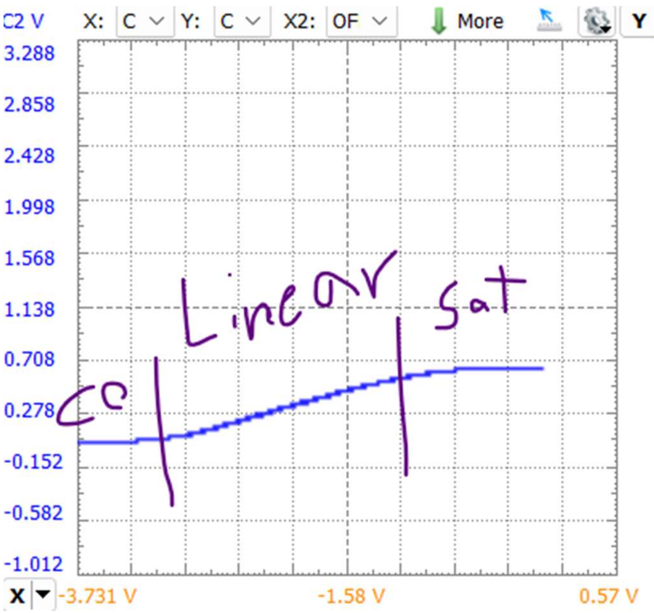


Part 3: Direct-Coupled Common-Source Amplifier (30 Points)

To operate the MOSFET in the active region, we choose the voltages/currents in a circuit so that the conditions for active operation are satisfied. We call this process ***biasing*** the circuit. The state of the circuit (node voltages and currents) is referred to as the ***operating point***.

We will attach a subscript "Q" to key parameters to emphasize that they are the voltages and currents at our operating point ($V_G = V_{GQ}$, $V_D = V_{DQ}$ and $I_D = I_{DQ}$).

I. Using the same circuit from Part 1, set the WaveGen back to the DC mode. Adjust V_G so that V_D is roughly halfway between the cutoff and triode limits. In other words, V_D is halfway between the limit for reaching the cutoff region, and halfway between reaching the triode region. Try it out yourself but what you will eventually find is that it is in the range of 2.5 V.
State your V_G regardless. **(5 Points)**

2.328V for VG

This is a good operating point for your amplifier since it:

a) Places V_G above the threshold voltage, allowing current to flow through I_D, and
b) Keeps V_G low enough to avoid a large current I_D from flowing through R_D, which would cause V_D to drop below $V_{Dsat} = V_G - V_T$, placing the transistor in the triode region.

II. Measure and record V_{DQ} and R_D and use them to calculate I_{DQ} at the operating point. Notice that you have to set V_G carefully to get the values of V_{DQ} and I_{DQ} you want. **(3 Points)**

VDQ = 2.5V, RD = 1k

III. Use hand calculations to verify that your MOSFET is operating in the active region based on your measured V_{GQ} and V_{DQ} values. **(3 Points)**

VD >= VG – VT → 2.5 >= (2.328 – 1.92) → 2.5 >= 0.48

IV. Temporarily swap your 2N7000 for a different one and repeat the same steps II and III using the same value for V_{GQ}. **(2 Points)**

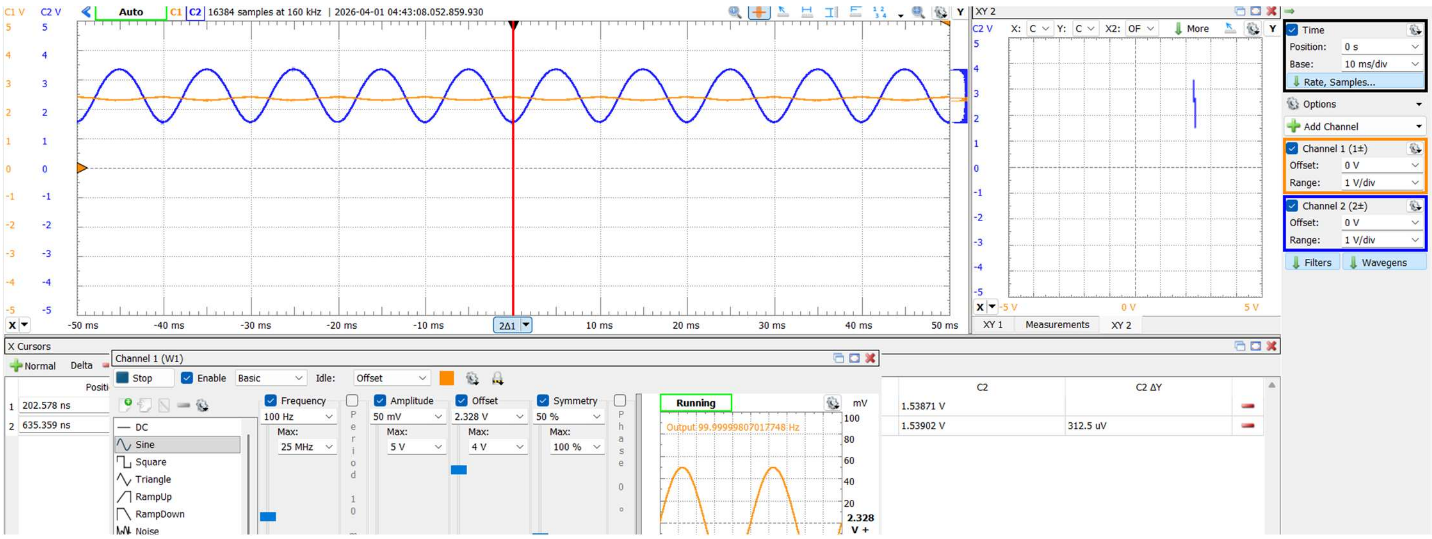
V. Was there a significant difference in the operating point? **(2 Points)**

* Use your original transistor for the rest of the experiment.

Yes, dropped VG to 2.295 to get 2.5V on VD

VI. Set the WaveGen to output a sine wave with 50 mV amplitude with DC offset equal to the operating-point value of V_G you just chose. Observe the output signal component of $V_D(t)$ on the scope and take a screenshot. **(5 Points)**

* If the output signal doesn't look sinusoidal, reduce the input signal amplitude or go back and choose an operating point further from the cutoff and triode-region limits.



VII. Measure the amplitudes of the input and output sinusoids v_o and v_i. Compute the voltage gain: $A_v = v_o / v_i$. Save the waveforms and show your calculations. **(5 Points)**

	Name	
C2	Amplitude	0.90519 V
C1	Amplitude	52.327 mV

0.90519/(52.327*10^-3) = 17.2987176792

* When you take measurements using the scope, always set the sensitivity and offsets so that the waveforms look big on the scope screen, otherwise you'll lose precision in the measurement.

VIII. Increase the input amplitude and note the distortion of the output waveform as it operates over the more obviously nonlinear portions of the VTC. Screenshot the distorted waveforms, don't calculate the gain. **(5 Points)**

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